

Scenario-Based Quiz: Retrieval-Augmented Generation (RAG)

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Date: _____ 16 March 2026 _____

Score: _____ / 20

Scenario 1: University Library Assistant

A university builds a **RAG chatbot** that answers questions from thousands of **digitized textbooks and research papers**. The system converts the documents into embeddings and stores them in a **vector database**.

1. When a student asks “*Explain supervised learning*”, what is the **first step** the RAG system performs?

- A. Generate the answer directly
 - B. Retrieve relevant document chunks
 - C. Train a new model
 - D. Summarize the entire database
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2. Why does the system convert textbook content into **embeddings**?

- A. To store images
 - B. To perform similarity search
 - C. To compress the database
 - D. To increase file size
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Scenario 2: Legal Compliance Chatbot

A law firm creates a chatbot that answers questions based on **regulatory documents**. Each document is divided into **small chunks before embedding**.

3. Why is chunking necessary in this system?

- A. To make documents visually attractive
 - B. To improve retrieval of relevant information**
 - C. To increase storage cost
 - D. To convert text to speech
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4. If chunks are **too large**, what problem might occur?

- A. Better accuracy always
 - B. Irrelevant information may be retrieved**
 - C. Vector database will stop working
 - D. The LLM cannot generate answers
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Scenario 3: Healthcare Knowledge Assistant

A hospital deploys a RAG system that retrieves medical guidelines and injects them into the LLM prompt before generating responses.

5. What is this step called?

- A. Data preprocessing
 - B. Context injection**
 - C. Data compression
 - D. Model training
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6. Why is context injection important?

- A. It helps the LLM generate answers using relevant knowledge**
 - B. It increases the dataset size
 - C. It replaces embeddings
 - D. It deletes irrelevant data
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Scenario 4: Customer Support Chatbot

A telecom company builds a RAG chatbot trained on **FAQs, product manuals, and troubleshooting guides**.

Users complain that sometimes the chatbot gives **incorrect answers even though the correct document exists**.

7. Which issue is most likely causing this problem?

- A. Poor retrieval quality
 - B. Too many GPUs
 - C. Large language model size
 - D. Internet speed
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8. Which evaluation metric would help diagnose the problem?

- A. Retrieval relevance
 - B. Screen resolution
 - C. Disk usage
 - D. Network latency
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Scenario 5: AI Research Assistant

An AI research lab uses a RAG system to summarize recent research papers. Engineers are experimenting with **different chunk sizes and retrieval strategies**.

9. What is the goal of testing different chunk sizes?

- A. Reduce the number of documents
 - B. Optimize retrieval quality and context relevance
 - C. Remove embeddings
 - D. Replace the LLM
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10. Which step in the RAG pipeline happens **after retrieval**?

- A. Tokenization of the dataset
 - B. Context injection into the prompt
 - C. Training the LLM
 - D. Deleting document chunks
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✅ Total Marks: 20

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